Supplemental Material for "Extracting Nonlinear Dynamical Response Functions from Time Evolution"

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I. DERIVATION OF EQUATIONS IN THE MAIN TEXT

A. Equation (1)

The perturbative response of a physical quantity $\langle Q_\alpha \rangle(t)$ to external fields $\{f_\beta(t)\}$ is generally written as

$$\langle Q_\alpha \rangle(t) = \sum_{n=0}^{\infty} \langle Q_\alpha \rangle^{(n)}(t), \quad \langle Q_\alpha \rangle^{(n)}(t) = \sum_{\{\beta_i\}} \int_{-\infty}^{\infty} \mathrm{d}t_1 \cdots \mathrm{d}t_n \, \chi^{(n)}_{\alpha\beta_1\cdots\beta_n}(t;t_1,\ldots,t_n) f_{\beta_1}(t_1)\cdots f_{\beta_n}(t_n). \tag{S1}$$

Here, $\chi^{(n)}_{\alpha\beta_1\cdots\beta_n}(t;t_1,\ldots,t_n)$ is the nth-order retarded response function, which is nonzero when $t \geq t_1 \geq t_2 \geq \cdots \geq t_n$ and vanishes otherwise because of causality. In addition, when the system is in a stationary state, the response function is invariant under time translation:

$$\chi^{(n)}_{\alpha\beta_1\cdots\beta_n}(t;t_1,\ldots,t_n) = \chi^{(n)}_{\alpha\beta_1\cdots\beta_n}(t+T;t_1+T,\ldots,t_n+T) \tag{S2}$$

for any T. Thus, the response function can be written as

$$\chi^{(n)}_{\alpha\beta_1\cdots\beta_n}(t;t_1,\ldots,t_n) = \chi^{(n)}_{\alpha\beta_1\cdots\beta_n}(0;t_1-t,\ldots,t_n-t) \equiv \chi^{(n)}_{\alpha\beta_1\cdots\beta_n}(\bar{t}_1,\ldots,\bar{t}_n), \tag{S3}$$

with $\bar{t}_i = t - t_i$. The Fourier transformation of $\chi^{(n)}_{\alpha\beta_1\cdots\beta_n}(t;t_1,\ldots,t_n)$ is defined by

$$\chi^{(n)}_{\alpha\beta_1\cdots\beta_n}(\omega;\omega_1,\ldots,\omega_n) = \int_{-\infty}^{\infty} \mathrm{d}t\mathrm{d}t_1\cdots\mathrm{d}t_n \, \mathrm{e}^{\mathrm{i}\omega t + \mathrm{i}\omega_1 t_1 + \cdots + \mathrm{i}\omega_n t_n} \chi^{(n)}_{\alpha\beta_1\cdots\beta_n}(t;t_1,\ldots,t_n), \tag{S4}$$

and for stationarity [Eq. (S2)], it reduces to

$$\begin{aligned}\chi^{(n)}_{\alpha\beta_1\cdots\beta_n}(\omega;-\omega_1,\ldots,-\omega_n) &= \int_{-\infty}^{\infty} \mathrm{d}t \, \mathrm{e}^{\mathrm{i}(\omega-\omega_1-\cdots-\omega_n)t} \int_{-\infty}^{\infty} \mathrm{d}\bar{t}_1\cdots\mathrm{d}\bar{t}_n \, \mathrm{e}^{\mathrm{i}\omega_1\bar{t}_1+\cdots+\mathrm{i}\omega_n\bar{t}_n} \chi^{(n)}_{\alpha\beta_1\cdots\beta_n}(\bar{t}_1,\ldots,\bar{t}_n) \\ &= 2\pi\delta(\omega_1+\cdots+\omega_n-\omega)\chi^{(n)}_{\alpha\beta_1\cdots\beta_n}(\omega_1,\ldots,\omega_n),\end{aligned} \tag{S5}$$

where

$$\chi^{(n)}_{\alpha\beta_1\cdots\beta_n}(\omega_1,\ldots,\omega_n) = \int_{-\infty}^{\infty} \mathrm{d}\bar{t}_1\cdots\mathrm{d}\bar{t}_n \, \mathrm{e}^{\mathrm{i}\omega_1\bar{t}_1+\cdots+\mathrm{i}\omega_n\bar{t}_n} \chi^{(n)}_{\alpha\beta_1\cdots\beta_n}(\bar{t}_1,\ldots,\bar{t}_n). \tag{S6}$$

Given the above, we obtain the expression for the nth-order frequency response as follows:

$$\begin{aligned}\langle Q_\alpha \rangle^{(n)}(\omega) &= \int_{-\infty}^{\infty} \mathrm{d}t \, \mathrm{e}^{\mathrm{i}\omega t} \langle Q_\alpha \rangle^{(n)}(t) \\ &= \sum_{\{\beta_i\}} \int_{-\infty}^{\infty} \mathrm{d}t \, \mathrm{e}^{\mathrm{i}\omega t} \int_{-\infty}^{\infty} \mathrm{d}t_1\cdots\mathrm{d}t_n \, \chi^{(n)}_{\alpha\beta_1\cdots\beta_n}(t;t_1,\ldots,t_n) f_{\beta_1}(t_1)\cdots f_{\beta_n}(t_n) \\ &= \sum_{\{\beta_i\}} \int_{-\infty}^{\infty} \mathrm{d}t \, \mathrm{e}^{\mathrm{i}\omega t} \int_{-\infty}^{\infty} \mathrm{d}t_1\cdots\mathrm{d}t_n \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega'\mathrm{d}\omega'_1\cdots\mathrm{d}\omega'_n}{(2\pi)^{n+1}} \, \mathrm{e}^{-\mathrm{i}\omega' t - \mathrm{i}\omega'_1 t_1 - \cdots - \mathrm{i}\omega'_n t_n} \chi^{(n)}_{\alpha\beta_1\cdots\beta_n}(\omega';\omega'_1,\ldots,\omega'_n) \\ &\quad \times \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega_1\cdots\mathrm{d}\omega_n}{(2\pi)^n} \, \mathrm{e}^{-\mathrm{i}\omega_1 t_1 - \cdots - \mathrm{i}\omega_n t_n} f_{\beta_1}(\omega_1)\cdots f_{\beta_n}(\omega_n) \\ &= \sum_{\{\beta_i\}} \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega_1\cdots\mathrm{d}\omega_n}{(2\pi)^n} \, \chi^{(n)}_{\alpha\beta_1\cdots\beta_n}(\omega;-\omega_1,\ldots,-\omega_n) f_{\beta_1}(\omega_1)\cdots f_{\beta_n}(\omega_n) \\ &= \sum_{\{\beta_i\}} \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega_1\cdots\mathrm{d}\omega_n}{(2\pi)^{n-1}} \, \delta(\omega_1+\cdots+\omega_n-\omega)\chi^{(n)}_{\alpha\beta_1\cdots\beta_n}(\omega_1,\ldots,\omega_n) f_{\beta_1}(\omega_1)\cdots f_{\beta_n}(\omega_n).\end{aligned} \tag{S7}$$

This is Eq. (1) in the main text.

B. Equations (2) and (4)

The $(n-1)$th functional derivative of $\langle Q_\alpha\rangle^{(n)}(\omega)$ in Eq. (S7) is

$$\frac{\delta^{n-1}\langle Q_\alpha\rangle^{(n)}(\omega)}{\delta f_{\alpha_{n-1}}(\omega_{n-1})\cdots\delta f_{\alpha_1}(\omega_1)} = \sum_{\beta_n}\frac{\bar{\chi}^{(n)}_{\alpha\alpha_1\cdots\alpha_{n-1}\beta_n}(\omega_1,\ldots,\omega_n)f_{\beta_n}(\omega_n)}{(2\pi)^{n-1}}, \tag{S8}$$

where $\bar{\chi}^{(n)}$ is the nth-order symmetric response function defined in Eq. (3). For simplicity, we consider the second-order response and assume that the indices run over two Cartesian directions, namely, $\alpha,\beta\in\{x,y\}$. In this case, we have

$$\frac{\delta\langle Q_x\rangle^{(2)}(\omega)}{\delta f_x(\omega_1)} = \sum_{\beta_2}\frac{\bar{\chi}^{(2)}_{xx\beta_2}(\omega_1,\omega_2)f_{\beta_2}(\omega_2)}{2\pi} = \frac{1}{2\pi}\Big[\bar{\chi}^{(2)}_{xxx}(\omega_1,\omega_2)f_x(\omega_2)+\bar{\chi}^{(2)}_{xxy}(\omega_1,\omega_2)f_y(\omega_2)\Big]. \tag{S9}$$

Thus, differentiating with respect to $f_{\beta_2}(\omega_2)$ yields

$$\frac{\partial}{\partial f_x(\omega_2)}\frac{\delta\langle Q_x\rangle^{(2)}(\omega)}{\delta f_x(\omega_1)} = \frac{1}{2\pi}\bar{\chi}^{(2)}_{xxx}(\omega_1,\omega_2), \quad \frac{\partial}{\partial f_y(\omega_2)}\frac{\delta\langle Q_x\rangle^{(2)}(\omega)}{\delta f_x(\omega_1)} = \frac{1}{2\pi}\bar{\chi}^{(2)}_{xxy}(\omega_1,\omega_2). \tag{S10}$$

By generalizing this result, we obtain Eq. (2):

$$\bar{\chi}^{(n)}_{\alpha\alpha_1\cdots\alpha_n}(\omega_1,\ldots,\omega_n) = (2\pi)^{n-1}\frac{\partial}{\partial f_{\alpha_n}(\omega_n)}\frac{\delta^{n-1}\langle Q_\alpha\rangle^{(n)}(\omega)}{\delta f_{\alpha_{n-1}}(\omega_{n-1})\cdots\delta f_{\alpha_1}(\omega_1)}. \tag{S11}$$

If we omit the indices, the summation in Eq. (S8) disappears, allowing us to obtain $\bar{\chi}^{(n)}(\omega_1,\ldots,\omega_n)$ by simply dividing Eq. (S8) by $f(\omega_n)$:

$$\bar{\chi}^{(n)}(\omega_1,\ldots,\omega_n) = \frac{(2\pi)^{n-1}}{f(\omega_n)}\frac{\delta^{n-1}\langle Q\rangle^{(n)}(\omega)}{\delta f(\omega_{n-1})\cdots\delta f(\omega_1)}, \tag{S12}$$

as shown in Eq. (4) in the main text.

C. Equation (9)

By directly calculating the first variation, we obtain Eq. (9) as follows:

$$\begin{aligned}
\delta_g\langle Q\rangle^{(2)}(\omega) &\equiv \langle Q\rangle^{(2)}_{f+\epsilon g}(\omega) - \langle Q\rangle^{(2)}_{f}(\omega)\\
&= \int_{-\infty}^{\infty}\frac{\mathrm{d}\omega_1\,\mathrm{d}\omega_2}{2\pi}\,\delta(\omega_1+\omega_2-\omega)\,\chi^{(2)}(\omega_1,\omega_2)[f(\omega_1)+\epsilon g(\omega_1)][f(\omega_2)+\epsilon g(\omega_2)]\\
&\quad -\int_{-\infty}^{\infty}\frac{\mathrm{d}\omega_1\,\mathrm{d}\omega_2}{2\pi}\,\delta(\omega_1+\omega_2-\omega)\,\chi^{(2)}(\omega_1,\omega_2)f(\omega_1)f(\omega_2)\\
&= \epsilon\int_{-\infty}^{\infty}\frac{\mathrm{d}\omega_1\,\mathrm{d}\omega_2}{2\pi}\,\delta(\omega_1+\omega_2-\omega)\,\chi^{(2)}(\omega_1,\omega_2)[f(\omega_1)g(\omega_2)+g(\omega_1)f(\omega_2)]+O(\epsilon^2)\\
&= \epsilon\int_{-\infty}^{\infty}\frac{\mathrm{d}\omega_1\,\mathrm{d}\omega_2}{2\pi}\,\delta(\omega_1+\omega_2-\omega)[\chi^{(2)}(\omega_2,\omega_1)f(\omega_2)g(\omega_1)+\chi^{(2)}(\omega_1,\omega_2)g(\omega_1)f(\omega_2)]+O(\epsilon^2)\\
&= \epsilon\int_{-\infty}^{\infty}\mathrm{d}\omega_1\,g(\omega_1)\int_{-\infty}^{\infty}\frac{\mathrm{d}\omega_2}{2\pi}\,\delta(\omega_1+\omega_2-\omega)[\chi^{(2)}(\omega_1,\omega_2)+\chi^{(2)}(\omega_2,\omega_1)]f(\omega_2)+O(\epsilon^2)\\
&= \epsilon\int_{-\infty}^{\infty}\mathrm{d}\omega_1\,g(\omega_1)\frac{1}{2\pi}\underbrace{[\chi^{(2)}(\omega_1,\omega-\omega_1)+\chi^{(2)}(\omega-\omega_1,\omega_1)]}_{=\bar{\chi}^{(2)}(\omega_1,\omega-\omega_1)}f(\omega-\omega_1)+O(\epsilon^2)\\
&= \epsilon\int_{-\infty}^{\infty}\mathrm{d}\omega_1\,g(\omega_1)\frac{\bar{\chi}^{(2)}(\omega_1,\omega-\omega_1)f(\omega-\omega_1)}{2\pi}+O(\epsilon^2).
\end{aligned} \tag{S13}$$

Note that $\chi^{(2)}$ appears in its symmetric form as $\bar{\chi}^{(2)}$ [see Eq. (3)]; this also holds for higher order responses.

II. RELATIONSHIP TO TWO-DIMENSIONAL COHERENT SPECTROSCOPY

In two-dimensional coherent spectroscopy (2DCS), two delta-function pulses are applied at time 0 and τ (> 0), and then a physical quantity Q is measured at time $t + \tau$ ($> t > 0$). For simplicity, here we omit the indices of the physical quantities and assume that $\langle Q \rangle^{(0)} = 0$. The two delta-function pulses can be written as

$$f(t) = F_0 \delta(t), \quad g(t) = F_\tau \delta(t - \tau). \tag{S14}$$

From Eq. (S1), we obtain the following expressions up to the third order:

$$\begin{aligned}
\langle Q \rangle_f (t+\tau) &\approx \int_{-\infty}^{\infty} \mathrm{d}t_1 \, \chi^{(1)}(t+\tau; t_1) f(t_1) + \int_{-\infty}^{\infty} \mathrm{d}t_1 \mathrm{d}t_2 \, \chi^{(2)}(t+\tau; t_1, t_2) f(t_1) f(t_2) \\
&\quad + \int_{-\infty}^{\infty} \mathrm{d}t_1 \mathrm{d}t_2 \mathrm{d}t_3 \, \chi^{(3)}(t+\tau; t_1, t_2, t_3) f(t_1) f(t_2) f(t_3) \\
&= \chi^{(1)}(t+\tau; 0) F_0 + \chi^{(2)}(t+\tau; 0, 0) F_0^2 + \chi^{(3)}(t+\tau; 0, 0, 0) F_0^3, \qquad \text{(S15)} \\
\langle Q \rangle_g (t+\tau) &\approx \int_{-\infty}^{\infty} \mathrm{d}t_1 \, \chi^{(1)}(t+\tau; t_1) g(t_1) + \int_{-\infty}^{\infty} \mathrm{d}t_1 \mathrm{d}t_2 \, \chi^{(2)}(t+\tau; t_1, t_2) g(t_1) g(t_2) \\
&\quad + \int_{-\infty}^{\infty} \mathrm{d}t_1 \mathrm{d}t_2 \mathrm{d}t_3 \, \chi^{(3)}(t+\tau; t_1, t_2, t_3) g(t_1) g(t_2) g(t_3) \\
&= \chi^{(1)}(t+\tau; \tau) F_\tau + \chi^{(2)}(t+\tau; \tau, \tau) F_\tau^2 + \chi^{(3)}(t+\tau; \tau, \tau, \tau) F_\tau^3, \qquad \text{(S16)} \\
\langle Q \rangle_{f+g} (t+\tau) &\approx \int_{-\infty}^{\infty} \mathrm{d}t_1 \, \chi^{(1)}(t+\tau; t_1) [f(t_1) + g(t_1)] + \int_{-\infty}^{\infty} \mathrm{d}t_1 \mathrm{d}t_2 \, \chi^{(2)}(t+\tau; t_1, t_2) [f(t_1) + g(t_1)][f(t_2) + g(t_2)] \\
&\quad + \int_{-\infty}^{\infty} \mathrm{d}t_1 \mathrm{d}t_2 \mathrm{d}t_3 \, \chi^{(3)}(t+\tau; t_1, t_2, t_3) [f(t_1) + g(t_1)][f(t_2) + g(t_2)][f(t_3) + g(t_3)] \\
&= \chi^{(1)}(t+\tau; 0) F_0 + \chi^{(1)}(t+\tau; \tau) F_\tau \\
&\quad + \chi^{(2)}(t+\tau; 0, 0) F_0^2 + \cancel{\chi^{(2)}(t+\tau; 0, \tau)} F_0 F_\tau + \chi^{(2)}(t+\tau; \tau, 0) F_\tau F_0 + \chi^{(2)}(t+\tau; \tau, \tau) F_\tau^2 \\
&\quad + \chi^{(3)}(t+\tau; 0, 0, 0) F_0^3 + \left[\cancel{\chi^{(3)}(t+\tau; 0, 0, \tau)} + \cancel{\chi^{(3)}(t+\tau; 0, \tau, 0)} + \chi^{(3)}(t+\tau; \tau, 0, 0) \right] F_0^2 F_\tau \\
&\quad + \left[\cancel{\chi^{(3)}(t+\tau; 0, \tau, \tau)} + \cancel{\chi^{(3)}(t+\tau; \tau, 0, \tau)} + \chi^{(3)}(t+\tau; \tau, \tau, 0) \right] F_0 F_\tau^2 + \chi^{(3)}(t+\tau; \tau, \tau, \tau) F_\tau^3. \qquad \text{(S17)}
\end{aligned}$$

Therefore,

$$\begin{aligned}
\langle Q \rangle_{\mathrm{NL}} (t+\tau) &\equiv \langle Q \rangle_{f+g} (t+\tau) - \langle Q \rangle_f (t+\tau) - \langle Q \rangle_g (t+\tau) \\
&\approx \chi^{(2)}(t+\tau; \tau, 0) F_\tau F_0 + \chi^{(3)}(t+\tau; \tau, 0, 0) F_0^2 F_\tau + \chi^{(3)}(t+\tau; \tau, \tau, 0) F_0 F_\tau^2 \\
&= \chi^{(2)}(t, t+\tau) F_\tau F_0 + \chi^{(3)}(t, t+\tau, t+\tau) F_0^2 F_\tau + \chi^{(3)}(t, t, t+\tau) F_0 F_\tau^2. \qquad \text{(S18)}
\end{aligned}$$

Here, we have used the causality condition, such as $\chi^{(2)}(t+\tau; 0, \tau) = 0$ for $\tau > 0$. We can numerically extract the second- and third-order response functions in Eq. (S18) by varying F_0 and F_τ. In the conventional 2DCS, the Fourier transformation with respect to t and τ are performed. For the second-order response, we find that

$$\tilde{\chi}^{(2)}(\omega_t, \omega_\tau) \equiv \int_{-\infty}^{\infty} \mathrm{d}t \mathrm{d}\tau \, \mathrm{e}^{\mathrm{i}\omega_t t + \mathrm{i}\omega_\tau \tau} \chi^{(2)}(t, t+\tau) = \int_{-\infty}^{\infty} \mathrm{d}\bar{t}_1 \mathrm{d}\bar{t}_2 \, \mathrm{e}^{\mathrm{i}(\omega_t - \omega_\tau)\bar{t}_1 + \mathrm{i}\omega_\tau \bar{t}_2} \chi^{(2)}(\bar{t}_1, \bar{t}_2) = \chi^{(2)}(\omega_t - \omega_\tau, \omega_\tau), \tag{S19}$$

which indicates that 2DCS can provide $\chi^{(2)}(\omega_1, \omega_2)$ in Eq. (S6) across the entire two-dimensional frequency space. Note that when $\tau < 0$, $\chi^{(2)}(t+\tau; \tau, 0) = \chi^{(2)}(t, t+\tau)$ vanishes, whereas the other second-order term, $\chi^{(2)}(t+\tau; 0, \tau) = \chi^{(2)}(t+\tau, t)$, becomes nonzero. In this case, the Fourier transformation yields:

$$\tilde{\chi}^{(2')}(\omega_t, \omega_\tau) \equiv \int_{-\infty}^{\infty} \mathrm{d}t \mathrm{d}\tau \, \mathrm{e}^{\mathrm{i}\omega_t t + \mathrm{i}\omega_\tau \tau} \chi^{(2)}(t+\tau, t) = \int_{-\infty}^{\infty} \mathrm{d}\bar{t}_1 \mathrm{d}\bar{t}_2 \, \mathrm{e}^{\mathrm{i}\omega_\tau \bar{t}_1 + \mathrm{i}(\omega_t - \omega_\tau)\bar{t}_2} \chi^{(2)}(\bar{t}_1, \bar{t}_2) = \chi^{(2)}(\omega_\tau, \omega_t - \omega_\tau). \tag{S20}$$

Thus, $\chi^{(2)}(t+\tau; \tau, 0)$ for $\tau > 0$ and $\chi^{(2)}(t+\tau; 0, \tau)$ for $\tau < 0$ contain equivalent information about $\chi^{(2)}(\omega_1, \omega_2)$.

In contrast, for the third-order response, it is shown that

$$\begin{aligned}
\tilde{\chi}^{(3,1)}(\omega_t,\omega_\tau) &\equiv \int_{-\infty}^{\infty} \mathrm{d}t\mathrm{d}\tau\, \mathrm{e}^{\mathrm{i}\omega_t t+\mathrm{i}\omega_\tau \tau}\chi^{(3)}(t,t+\tau,t+\tau) \\
&= \int_{-\infty}^{\infty} \mathrm{d}t\mathrm{d}\tau\mathrm{d}\tau'\, \mathrm{e}^{\mathrm{i}\omega_t t+\mathrm{i}\omega_\tau \tau}\delta(\tau'-\tau)\chi^{(3)}(t,t+\tau,t+\tau') \\
&= \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega'}{2\pi}\int_{-\infty}^{\infty} \mathrm{d}t\mathrm{d}\tau\mathrm{d}\tau'\, \mathrm{e}^{\mathrm{i}\omega_t t+\mathrm{i}\omega_\tau \tau+\mathrm{i}\omega'(\tau'-\tau)}\chi^{(3)}(t,t+\tau,t+\tau') \\
&= \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega'}{2\pi}\int_{-\infty}^{\infty} \mathrm{d}\bar{t}_1\mathrm{d}\bar{t}_2\mathrm{d}\bar{t}_3\, \mathrm{e}^{\mathrm{i}(\omega_t-\omega_\tau)\bar{t}_1+\mathrm{i}(\omega_\tau-\omega')\bar{t}_2+\mathrm{i}\omega'\bar{t}_3}\chi^{(3)}(\bar{t}_1,\bar{t}_2,\bar{t}_3) \\
&= \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega'}{2\pi}\,\chi^{(3)}(\omega_t-\omega_\tau,\omega_\tau-\omega',\omega'),
\end{aligned} \tag{S21}$$

and

$$\begin{aligned}
\tilde{\chi}^{(3,2)}(\omega_t,\omega_\tau) &\equiv \int_{-\infty}^{\infty} \mathrm{d}t\mathrm{d}\tau\, \mathrm{e}^{\mathrm{i}\omega_t t+\mathrm{i}\omega_\tau \tau}\chi^{(3)}(t,t,t+\tau) \\
&= \int_{-\infty}^{\infty} \mathrm{d}t\mathrm{d}t'\mathrm{d}\tau\, \mathrm{e}^{\mathrm{i}\omega_t t+\mathrm{i}\omega_\tau \tau}\delta(t'-t)\chi^{(3)}(t,t',t'+\tau) \\
&= \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega'}{2\pi}\int_{-\infty}^{\infty} \mathrm{d}t\mathrm{d}t'\mathrm{d}\tau\, \mathrm{e}^{\mathrm{i}\omega_t t+\mathrm{i}\omega_\tau \tau+\mathrm{i}\omega'(t'-t)}\chi^{(3)}(t,t',t'+\tau) \\
&= \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega'}{2\pi}\int_{-\infty}^{\infty} \mathrm{d}\bar{t}_1\mathrm{d}\bar{t}_2\mathrm{d}\bar{t}_3\, \mathrm{e}^{\mathrm{i}(\omega_t-\omega')\bar{t}_1+\mathrm{i}(\omega'-\omega_\tau)\bar{t}_2+\mathrm{i}\omega_\tau\bar{t}_3}\chi^{(3)}(\bar{t}_1,\bar{t}_2,\bar{t}_3) \\
&= \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega'}{2\pi}\,\chi^{(3)}(\omega_t-\omega',\omega'-\omega_\tau,\omega_\tau).
\end{aligned} \tag{S22}$$

Equations (S21) and (S22) indicate that, for the third and higher orders, only the *partial* response functions can be obtained through 2DCS. Although the *full* nth-order response functions can, in principle, be retrieved by n-dimensional spectroscopy, the computational cost increases significantly because of the need to sweep the time delays between n delta-function pulses.

III. RESPONSE FUNCTIONS IN QUANTUM SYSTEMS

Suppose that the Hamiltonian of a quantum system at time t is given by

$$\mathcal{H}(t) = \mathcal{H}_0 + \mathcal{V}(t), \quad \mathcal{V}(t) = -\sum_\beta f_\beta(t) Q_\beta, \tag{S23}$$

where Q_β is a physical quantity labeled by β, and $f_\beta(t)$ is an external field linearly coupled to Q_β. The interaction-picture operator of Q_α is defined by

$$\hat{Q}_\alpha(t) = U_0(t_0,t) Q_\alpha U_0(t,t_0), \quad U_0(t,t_0) = \mathrm{e}^{-\mathrm{i}\mathcal{H}_0\cdot(t-t_0)}. \tag{S24}$$

The time evolution of the many-body density matrix $\rho(t)$ is governed by the von Neumann equation,

$$\mathrm{i}\partial_t \rho(t) = [\mathcal{H}(t), \rho(t)], \tag{S25}$$

and its solution can be formally written as

$$\rho(t) = U(t,t_0)\rho(t_0)U(t_0,t), \quad U(t,t_0) = \mathcal{T}_{C(t,t_0)} \exp\left[-\mathrm{i}\int_{t_0}^{t} \mathrm{d}\bar{t}\,\mathcal{H}(\bar{t})\right], \tag{S26}$$

where $\mathcal{T}_{C(t,t_0)}$ represents time-ordering along the contour from time t_0 to t. Here, we introduce the S-matrix:

$$S(t,t_0) = U_0(t,t_0)^\dagger U(t,t_0) = \mathcal{T}_{C(t,t_0)} \exp\left[-\mathrm{i}\int_{t_0}^{t} \mathrm{d}\bar{t}\,\hat{\mathcal{V}}(\bar{t})\right], \tag{S27}$$

and define the density matrix in the interaction picture as

$$\hat{\rho}(t) = S(t, t_0)\rho(t_0)S(t_0, t), \tag{S28}$$

which satisfies the following equation:

$$\mathrm{i}\partial_t \hat{\rho}(t) = [\hat{\mathcal{V}}(t), \hat{\rho}(t)]. \tag{S29}$$

The successive integration of Eq. (S29) yields

$$\hat{\rho}(t) = \rho_0 + \sum_{n=1}^{\infty} (-\mathrm{i})^n \int_{t_0}^{t} \mathrm{d}t_1 \int_{t_0}^{t_1} \mathrm{d}t_2 \cdots \int_{t_0}^{t_{n-1}} \mathrm{d}t_n \, [\hat{\mathcal{V}}(t_1), [\hat{\mathcal{V}}(t_2), \cdots [\hat{\mathcal{V}}(t_n), \rho_0] \cdots]], \tag{S30}$$

where $\rho_0 = \hat{\rho}(t_0) = \rho(t_0)$. Then, we obtain

$$\langle Q_\alpha \rangle(t) = \mathrm{Tr}[\hat{\rho}(t)\hat{Q}_\alpha(t)] = \langle Q_\alpha \rangle_0 + \sum_{n=1}^{\infty} \langle Q_\alpha \rangle^{(n)}(t), \tag{S31}$$

$$\langle Q_\alpha \rangle^{(n)}(t) = (-\mathrm{i})^n \int_{t_0}^{t} \mathrm{d}t_1 \int_{t_0}^{t_1} \mathrm{d}t_2 \cdots \int_{t_0}^{t_{n-1}} \mathrm{d}t_n \left\langle [\cdots [[\hat{Q}_\alpha(t), \hat{\mathcal{V}}(t_1)], \hat{\mathcal{V}}(t_2)], \cdots \hat{\mathcal{V}}(t_n)] \right\rangle_0, \tag{S32}$$

with $\langle \bullet \rangle_0 = \langle \bullet \rangle^{(0)} = \mathrm{Tr}[\rho_0 \bullet]$, where we have iteratively used the formula: $\mathrm{Tr}([A, B]C) = \mathrm{Tr}(B[C, A])$. By taking the initial time to be $t_0 \to -\infty$ and substituting Eq. (S23), Eq. (S32) can be written as

$$\langle Q_\alpha \rangle^{(n)}(t) = \sum_{\{\beta_i\}} \int_{-\infty}^{\infty} \mathrm{d}t_1 \cdots \mathrm{d}t_n \, \chi^{(n)}_{\alpha\beta_1\cdots\beta_n}(t; t_1, \ldots, t_n) f_{\beta_1}(t_1) \cdots f_{\beta_n}(t_n), \tag{S33}$$

where $\chi^{(n)}_{\alpha\beta_1\cdots\beta_n}(t; t_1, \ldots, t_n)$ is the nth-order response function defined by

$$\chi^{(n)}_{\alpha\beta_1\cdots\beta_n}(t; t_1, \ldots, t_n) = \mathrm{i}^n \Theta(t - t_1)\Theta(t_1 - t_2) \cdots \Theta(t_{n-1} - t_n) \left\langle [\cdots [[\hat{Q}_\alpha(t), \hat{Q}_{\beta_1}(t_1)], \hat{Q}_{\beta_2}(t_2)], \cdots \hat{Q}_{\beta_n}(t_n)] \right\rangle_0. \tag{S34}$$

Here, Θ denotes the unit step function, ensuring causality.

Let us consider a free fermion system whose Hamiltonian $\mathcal{H}_0$ is given by

$$\mathcal{H}_0 = \sum_{ij} h_{ij} c_i^\dagger c_j = \sum_{\nu} \varepsilon_\nu a_\nu^\dagger a_\nu, \tag{S35}$$

where $c_i^\dagger$ and $a_\nu^\dagger$ are the creation operators for fermions, and ε_ν denotes the one-body energy. In the following, we derive the expressions for the first-, second-, and third-order response of a physical quantity,

$$Q_\alpha = \sum_{ij} Q_\alpha^{ij} c_i^\dagger c_j = \sum_{\mu\nu} Q_\alpha^{\mu\nu} a_\mu^\dagger a_\nu, \tag{S36}$$

assuming that Q_α is time-independent in the Schrödinger picture. Using the commutation relations:

$$[a_\mu^\dagger a_\nu, a_{\mu'}^\dagger a_{\nu'}] = \delta_{\nu\mu'} a_\mu^\dagger a_{\nu'} - \delta_{\mu\nu'} a_{\mu'}^\dagger a_\nu, \tag{S37}$$

$$[a_\mu^\dagger a_\nu, \mathcal{H}_0] = (\varepsilon_\nu - \varepsilon_\mu) a_\mu^\dagger a_\nu, \tag{S38}$$

we obtain the expression for the first-order response,

$$\chi^{(1)}_{\alpha\beta}(t; t_1) = \mathrm{i}\Theta(t - t_1) \sum_{\mu\nu} Q_\alpha^{\mu\nu} Q_\beta^{\nu\mu} \mathrm{e}^{-\mathrm{i}(\varepsilon_\nu - \varepsilon_\mu)(t - t_1)} (n_\mu - n_\nu) = \chi^{(1)}_{\alpha\beta}(\bar{t}_1), \tag{S39}$$

$$\chi^{(1)}_{\alpha\beta}(\omega) = \int_{-\infty}^{\infty} \mathrm{d}\bar{t}_1 \, \mathrm{e}^{\mathrm{i}\omega^+ \bar{t}_1} \chi^{(1)}_{\alpha\beta}(\bar{t}_1) = \sum_{\mu\nu} Q_\alpha^{\mu\nu} Q_\beta^{\nu\mu} \frac{n_\nu - n_\mu}{\omega^+ - (\varepsilon_\nu - \varepsilon_\mu)}, \tag{S40}$$

where $n_\nu = \langle a_\nu^\dagger a_\nu \rangle_0$ corresponds to the Fermi–Dirac distribution function, and ω^+ denotes ω with a small positive imaginary part η, i.e., $\omega^+ = \omega + \mathrm{i}\eta$. Similarly, the second-order response is derived as follows:

$$\chi^{(2)}_{\alpha\beta_1\beta_2}(\omega_1, \omega_2) = \sum_{\mu_i \nu_i} \left[\frac{Q_\alpha^{\mu_0\nu_0} Q_{\beta_1}^{\nu_0\nu_1} Q_{\beta_2}^{\nu_1\mu_0} (n_{\mu_0} - n_{\nu_1})}{(\omega_1^+ + \omega_2^+ - \varepsilon_{\nu_0} + \varepsilon_{\mu_0})(\omega_2^+ - \varepsilon_{\nu_1} + \varepsilon_{\mu_0})} + \frac{Q_\alpha^{\mu_0\nu_0} Q_{\beta_2}^{\nu_0\mu_1} Q_{\beta_1}^{\mu_1\mu_0} (n_{\nu_0} - n_{\mu_1})}{(\omega_1^+ + \omega_2^+ - \varepsilon_{\nu_0} + \varepsilon_{\mu_0})(\omega_2^+ - \varepsilon_{\nu_0} + \varepsilon_{\mu_1})} \right], \tag{S41}$$

and the third-order response can be expressed as:

$$\chi^{(3)}_{\alpha\beta_1\beta_2\beta_3}(\omega_1,\omega_2,\omega_3) = \sum_{\mu_i\nu_i}\Bigg[\frac{Q^{\mu_0\nu_0}_{\alpha}Q^{\nu_0\nu_1}_{\beta_1}Q^{\nu_1\nu_2}_{\beta_2}Q^{\nu_2\mu_0}_{\beta_3}(n_{\nu_2}-n_{\mu_0})}{(\omega_1^+ + \omega_2^+ + \omega_3^+ - \varepsilon_{\nu_0} + \varepsilon_{\mu_0})(\omega_2^+ + \omega_3^+ - \varepsilon_{\nu_1} + \varepsilon_{\mu_0})(\omega_3^+ - \varepsilon_{\nu_2} + \varepsilon_{\mu_0})} + \frac{Q^{\mu_0\nu_0}_{\alpha}Q^{\nu_0\nu_1}_{\beta_1}Q^{\nu_1\mu_2}_{\beta_3}Q^{\mu_2\mu_0}_{\beta_2}(n_{\mu_2}-n_{\nu_1})}{(\omega_1^+ + \omega_2^+ + \omega_3^+ - \varepsilon_{\nu_0} + \varepsilon_{\mu_0})(\omega_2^+ + \omega_3^+ - \varepsilon_{\nu_1} + \varepsilon_{\mu_0})(\omega_3^+ - \varepsilon_{\nu_1} + \varepsilon_{\mu_2})} + \frac{Q^{\mu_0\nu_0}_{\alpha}Q^{\nu_0\nu_2}_{\beta_2}Q^{\nu_2\mu_1}_{\beta_3}Q^{\mu_1\mu_0}_{\beta_1}(n_{\mu_1}-n_{\nu_2})}{(\omega_1^+ + \omega_2^+ + \omega_3^+ - \varepsilon_{\nu_0} + \varepsilon_{\mu_0})(\omega_2^+ + \omega_3^+ - \varepsilon_{\nu_0} + \varepsilon_{\mu_1})(\omega_3^+ - \varepsilon_{\nu_2} + \varepsilon_{\mu_1})} + \frac{Q^{\mu_0\nu_0}_{\alpha}Q^{\nu_0\mu_2}_{\beta_3}Q^{\mu_2\mu_1}_{\beta_2}Q^{\mu_1\mu_0}_{\beta_1}(n_{\nu_0}-n_{\mu_2})}{(\omega_1^+ + \omega_2^+ + \omega_3^+ - \varepsilon_{\nu_0} + \varepsilon_{\mu_0})(\omega_2^+ + \omega_3^+ - \varepsilon_{\nu_0} + \varepsilon_{\mu_1})(\omega_3^+ - \varepsilon_{\nu_0} + \varepsilon_{\mu_2})}\Bigg]. \tag{S42}$$

Note that these higher-order response functions need to be symmetrized as follows:

$$\bar{\chi}^{(2)}_{\alpha\beta_1\beta_2}(\omega_1,\omega_2) = \sum_{\sigma\in\mathfrak{S}_2}\chi^{(2)}_{\alpha\beta_{\sigma(1)}\beta_{\sigma(2)}}(\omega_{\sigma(1)},\omega_{\sigma(2)}) = \chi^{(2)}_{\alpha\beta_1\beta_2}(\omega_1,\omega_2) + \chi^{(2)}_{\alpha\beta_2\beta_1}(\omega_2,\omega_1), \tag{S43}$$

$$\bar{\chi}^{(3)}_{\alpha\beta_1\beta_2\beta_3}(\omega_1,\omega_2,\omega_3) = \sum_{\sigma\in\mathfrak{S}_3}\chi^{(3)}_{\alpha\beta_{\sigma(1)}\beta_{\sigma(2)}\beta_{\sigma(3)}}(\omega_{\sigma(1)},\omega_{\sigma(2)},\omega_{\sigma(3)}) = (6\text{ terms from all permutations}). \tag{S44}$$

IV. NONLINEAR OPTICAL RESPONSES IN THE VELOCITY GAUGE

When introducing the vector potential $\boldsymbol{A}$ through Peierls substitution, the Hamiltonian of a free fermion system takes the following form:

$$\mathcal{H}[\boldsymbol{A}] = \sum_{\boldsymbol{k}}\sum_{ij}h_{ij}(\boldsymbol{k}-\boldsymbol{A})c^{\dagger}_{\boldsymbol{k}i}c_{\boldsymbol{k}j}, \tag{S45}$$

where $\boldsymbol{k}$ denotes the crystal momentum, and i and j are indices representing internal degrees of freedom, such as spin, orbital, and sublattice. This Hamiltonian describes a nonlinear coupling between the vector potential $\boldsymbol{A}$ and the electric current operator $\boldsymbol{J} = -\partial\mathcal{H}[\boldsymbol{A}]/\partial\boldsymbol{A}$. In this case, the results from the previous section cannot be directly applied because the current operator $\boldsymbol{J}$ itself depends on the external field $\boldsymbol{A}$. Therefore, $\boldsymbol{J}$ should be expanded in terms of $\boldsymbol{A}$ as follows:

$$J_\alpha = -\frac{\partial\mathcal{H}[\boldsymbol{A}]}{\partial A_\alpha} = \sum_{n=0}^{\infty}\sum_{\{\beta_i\}}J^{(n)}_{\alpha\beta_1\cdots\beta_n}A_{\beta_1}\cdots A_{\beta_n}, \quad J^{(n)}_{\beta_0\beta_1\cdots\beta_n} = \frac{(-1)^n}{n!}\sum_{\boldsymbol{k}ij}\partial_{\beta_0}\partial_{\beta_1}\cdots\partial_{\beta_n}h_{ij}(\boldsymbol{k})c^{\dagger}_{\boldsymbol{k}i}c_{\boldsymbol{k}j}, \tag{S46}$$

where $\partial_\beta = \partial/\partial k_\beta$. Note that $J^{(n)}_{\beta_0\beta_1\cdots\beta_n}$ is invariant under any permutation of its indices. By using this expansion, the Hamiltonian can be expressed as

$$\mathcal{H}[\boldsymbol{A}] = \mathcal{H}[0] - \sum_{n=0}^{\infty}\sum_{\{\beta_i\}}\frac{A_{\beta_0}\cdots A_{\beta_n}}{n+1}J^{(n)}_{\beta_0\cdots\beta_n}. \tag{S47}$$

Regarding the second term as a coupling term with $\boldsymbol{A}(t)$, we consider the response of the electric current density at time t:

$$\langle J_\alpha\rangle(t) = \frac{1}{V}\sum_{n=0}^{\infty}\sum_{\{\beta_i\}}\langle J^{(n)}_{\alpha\beta_1\cdots\beta_n}\rangle(t)A_{\beta_1}(t)\cdots A_{\beta_n}(t), \tag{S48}$$

where V represents the volume of the system. Since many types of physical quantities $\{J^{(n)}_{\alpha\beta_1\cdots\beta_n}\}$ and external fields $\{A_{\beta_0}\cdots A_{\beta_n}\}$ appear, we introduce the following notation for the nth-order response function:

$$\begin{aligned}&\langle\langle Q_\alpha(t); Q_{\beta_1}(t_1),\ldots,Q_{\beta_n}(t_n)\rangle\rangle = \langle\langle Q_\alpha; Q_{\beta_1},\ldots,Q_{\beta_n}\rangle\rangle(\bar{t}_1,\ldots,\bar{t}_n)\\ &= \mathrm{i}^n\Theta(t-t_1)\Theta(t_1-t_2)\cdots\Theta(t_{n-1}-t_n)\big\langle[\cdots[[\hat{Q}_\alpha(t),\hat{Q}_{\beta_1}(t_1)],\hat{Q}_{\beta_2}(t_2)],\cdots\hat{Q}_{\beta_n}(t_n)]\big\rangle_0\end{aligned} \tag{S49}$$

in the time domain, and

$$\langle\langle Q_\alpha; Q_{\beta_1},\ldots,Q_{\beta_n}\rangle\rangle(\omega_1,\ldots,\omega_n) = \int_{-\infty}^{\infty}\mathrm{d}\bar{t}_1\cdots\mathrm{d}\bar{t}_n\,\mathrm{e}^{\mathrm{i}\omega_1\bar{t}_1+\cdots+\mathrm{i}\omega_n\bar{t}_n}\langle\langle Q_\alpha; Q_{\beta_1},\ldots,Q_{\beta_n}\rangle\rangle(\bar{t}_1,\ldots,\bar{t}_n) \tag{S50}$$

in the frequency domain.

The first-order response is given by

$$\langle J_\alpha \rangle^{(1)}(t) = \frac{1}{V}\sum_{\beta_1}\left[\langle J^{(1)}_{\alpha\beta_1}\rangle_0 A_{\beta_1}(t) + \int_{-\infty}^{\infty} \mathrm{d}t_1 \, \langle\langle J^{(0)}_\alpha(t); J^{(0)}_{\beta_1}(t_1)\rangle\rangle A_{\beta_1}(t_1)\right]. \tag{S51}$$

Here, the first term represents the "zeroth-order" response of the diamagnetic current $J^{(1)}_{\alpha\beta_1}A_{\beta_1}$, and the second term represents the first-order response of the paramagnetic current $J^{(0)}_\alpha$ to the vector potential $\boldsymbol{A}$. The Fourier transformation of Eq. (S51) yields

$$\langle J_\alpha \rangle^{(1)}(\omega) = \sum_{\beta_1}\chi^{(1)}_{\alpha\beta_1}(\omega)A_{\beta_1}(\omega), \quad V\chi^{(1)}_{\alpha\beta_1}(\omega) = \langle J^{(1)}_{\alpha\beta_1}\rangle_0 + \langle\langle J^{(0)}_\alpha; J^{(0)}_{\beta_1}\rangle\rangle(\omega). \tag{S52}$$

The second-order response consists of the following four terms:

$$\begin{aligned}\langle J_\alpha \rangle^{(2)}(t) = \frac{1}{V}\sum_{\beta_1\beta_2}\Bigg[&\langle J^{(2)}_{\alpha\beta_1\beta_2}\rangle_0 A_{\beta_1}(t)A_{\beta_2}(t) + \int_{-\infty}^{\infty}\mathrm{d}t_1\mathrm{d}t_2\,\langle\langle J^{(0)}_\alpha(t); J^{(0)}_{\beta_1}(t_1), J^{(0)}_{\beta_2}(t_2)\rangle\rangle A_{\beta_1}(t_1)A_{\beta_2}(t_2)\\ &+\int_{-\infty}^{\infty}\mathrm{d}t_1\,\langle\langle J^{(0)}_\alpha(t); J^{(1)}_{\beta_1\beta_2}(t_1)\rangle\rangle\frac{A_{\beta_1}(t_1)A_{\beta_2}(t_2)}{2} + \int_{-\infty}^{\infty}\mathrm{d}t_1\,\langle\langle J^{(1)}_{\alpha\beta_1}(t); J^{(0)}_{\beta_2}(t_1)\rangle\rangle A_{\beta_1}(t)A_{\beta_2}(t_1)\Bigg].\end{aligned} \tag{S53}$$

Here, the first term is the zeroth-order response of $J^{(2)}_{\alpha\beta_1\beta_2}A_{\beta_1}A_{\beta_2}$; the second term represents the second-order response of $J^{(0)}_\alpha$ to $\boldsymbol{A}$; the third term denotes $J^{(0)}_\alpha$ induced by $J^{(1)}_{\beta_1\beta_2}A_{\beta_1}A_{\beta_2}/2$; and the fourth term corresponds to $J^{(1)}_{\alpha\beta_1}A_{\beta_1}$ induced by $J^{(0)}_{\beta_2}A_{\beta_2}$. By performing the Fourier transformation, we obtain

$$\langle J_\alpha \rangle^{(2)}(\omega) = \sum_{\beta_1\beta_2}\int_{-\infty}^{\infty}\frac{\mathrm{d}\omega_1\mathrm{d}\omega_2}{2\pi}\,\delta(\omega_1+\omega_2-\omega)\chi^{(2)}_{\alpha\beta_1\beta_2}(\omega_1,\omega_2)A_{\beta_1}(\omega_1)A_{\beta_2}(\omega_2), \tag{S54}$$

where

$$V\chi^{(2)}_{\alpha\beta_1\beta_2}(\omega_1,\omega_2) = \langle J^{(2)}_{\alpha\beta_1\beta_2}\rangle_0 + \langle\langle J^{(0)}_\alpha; J^{(0)}_{\beta_1}, J^{(0)}_{\beta_2}\rangle\rangle(\omega_1,\omega_2) + \frac{1}{2}\langle\langle J^{(0)}_\alpha; J^{(1)}_{\beta_1\beta_2}\rangle\rangle(\omega_1+\omega_2) + \langle\langle J^{(1)}_{\alpha\beta_1}; J^{(0)}_{\beta_2}\rangle\rangle(\omega_2). \tag{S55}$$

Similarly, the third-order response is expressed as

$$\langle J_\alpha \rangle^{(3)}(\omega) = \sum_{\beta_1\beta_2\beta_3}\int_{-\infty}^{\infty}\frac{\mathrm{d}\omega_1\mathrm{d}\omega_2\mathrm{d}\omega_3}{(2\pi)^2}\,\delta(\omega_1+\omega_2+\omega_3-\omega)\chi^{(3)}_{\alpha\beta_1\beta_2\beta_3}(\omega_1,\omega_2,\omega_3)A_{\beta_1}(\omega_1)A_{\beta_2}(\omega_2)A_{\beta_3}(\omega_3), \tag{S56}$$

where

$$\begin{aligned}&V\chi^{(3)}_{\alpha\beta_1\beta_2\beta_3}(\omega_1,\omega_2,\omega_3)\\ &= \langle J^{(3)}_{\alpha\beta_1\beta_2\beta_3}\rangle_0 + \langle\langle J^{(0)}_\alpha; J^{(0)}_{\beta_1}, J^{(0)}_{\beta_2}, J^{(0)}_{\beta_3}\rangle\rangle(\omega_1,\omega_2,\omega_3)\\ &\quad+\frac{1}{2}\langle\langle J^{(0)}_\alpha; J^{(0)}_{\beta_1}, J^{(1)}_{\beta_2\beta_3}\rangle\rangle(\omega_1,\omega_2+\omega_3) + \frac{1}{2}\langle\langle J^{(0)}_\alpha; J^{(1)}_{\beta_1\beta_2}, J^{(0)}_{\beta_3}\rangle\rangle(\omega_1+\omega_2,\omega_3) + \langle\langle J^{(1)}_{\alpha\beta_1}; J^{(0)}_{\beta_2}, J^{(0)}_{\beta_3}\rangle\rangle(\omega_2,\omega_3)\\ &\quad+\frac{1}{3}\langle\langle J^{(0)}_\alpha; J^{(2)}_{\beta_1\beta_2\beta_3}\rangle\rangle(\omega_1+\omega_2+\omega_3) + \frac{1}{2}\langle\langle J^{(1)}_{\alpha\beta_1}; J^{(1)}_{\beta_2\beta_3}\rangle\rangle(\omega_2+\omega_3) + \langle\langle J^{(2)}_{\alpha\beta_1\beta_2}; J^{(0)}_{\beta_3}\rangle\rangle(\omega_3).\end{aligned} \tag{S57}$$

We can calculate each term in Eqs. (S55) and (S57) by replacing $\{Q_\beta\}$ in Eqs. (S40)–(S42) with the corresponding current operators $\{J^{(n)}_{\beta_0\cdots\beta_n}\}$. Additionally, these second- and third-order response functions must be symmetrized in accordance with Eqs. (S43) and (S44).

V. HARTREE–FOCK ANALYSIS OF THE RICE–MELE–HUBBARD MODEL

In the Hartree–Fock (HF) approximation, the electron–electron interaction term in the Rice–Mele–Hubbard model [Eq. (A1) in the main text] is decoupled as

$$\begin{aligned}c^\dagger_{j\uparrow}c_{j\uparrow}c^\dagger_{j\downarrow}c_{j\downarrow} \approx\ & c^\dagger_{j\uparrow}c_{j\uparrow}\langle c^\dagger_{j\downarrow}c_{j\downarrow}\rangle_0 + \langle c^\dagger_{j\uparrow}c_{j\uparrow}\rangle_0 c^\dagger_{j\downarrow}c_{j\downarrow} - \langle c^\dagger_{j\uparrow}c_{j\uparrow}\rangle_0\langle c^\dagger_{j\downarrow}c_{j\downarrow}\rangle_0\\ &- c^\dagger_{j\uparrow}c_{j\downarrow}\langle c^\dagger_{j\downarrow}c_{j\uparrow}\rangle_0 - \langle c^\dagger_{j\uparrow}c_{j\downarrow}\rangle_0 c^\dagger_{j\downarrow}c_{j\uparrow} + \langle c^\dagger_{j\uparrow}c_{j\downarrow}\rangle_0\langle c^\dagger_{j\downarrow}c_{j\uparrow}\rangle_0.\end{aligned} \tag{S58}$$

Here, the Fock terms, $\langle c^\dagger_{j\downarrow} c_{j\uparrow}\rangle_0$ and $\langle c^\dagger_{j\uparrow} c_{j\downarrow}\rangle_0$, serve only to restore spin rotational symmetry, and thus these can be omitted without loss of generality. The resulting one-body Hamiltonian takes the form:

$$\mathcal{H}^{\mathrm{HF}}_{\mathrm{RMH}} = \mathcal{H}^{\mathrm{HF},\uparrow}_{\mathrm{RMH}} + \mathcal{H}^{\mathrm{HF},\downarrow}_{\mathrm{RMH}} + U\sum_j \left(\frac{1}{4} - \langle n_{j\uparrow}\rangle_0 \langle n_{j\downarrow}\rangle_0\right), \tag{S59}$$

$$\mathcal{H}^{\mathrm{HF},\uparrow}_{\mathrm{RMH}} = -\sum_j \frac{h_x - (-1)^j h_y}{2}\left(c^\dagger_{j+1,\uparrow} c_{j\uparrow} + \mathrm{H.c.}\right) - h_z \sum_j (-1)^j n_{j\uparrow} + U\sum_j \left(\langle n_{j\downarrow}\rangle_0 - \frac{1}{2}\right) n_{j\uparrow}, \tag{S60}$$

$$\mathcal{H}^{\mathrm{HF},\downarrow}_{\mathrm{RMH}} = -\sum_j \frac{h_x - (-1)^j h_y}{2}\left(c^\dagger_{j+1,\downarrow} c_{j\downarrow} + \mathrm{H.c.}\right) - h_z \sum_j (-1)^j n_{j\downarrow} + U\sum_j \left(\langle n_{j\uparrow}\rangle_0 - \frac{1}{2}\right) n_{j\downarrow}, \tag{S61}$$

where $n_{js} = c^\dagger_{js} c_{js}$ denotes the number operator for an electron with real spin s at site j. Assuming the two-sublattice structure, we obtain the Fourier transform of $\mathcal{H}^{\mathrm{HF}}_{\mathrm{RMH}}$ as

$$\mathcal{H}^{\mathrm{HF}}_{\mathrm{RMH}} = \sum_k \tilde{\Psi}^\dagger_k \tilde{H}_k \tilde{\Psi}_k + U\sum_j \left(\frac{1}{4} - \langle n_{j\uparrow}\rangle_0 \langle n_{j\downarrow}\rangle_0\right), \tag{S62}$$

$$\begin{aligned}\tilde{H}_k = &-\left(\frac{U}{2} - \frac{U}{2}\langle\tau^\downarrow_0\rangle_0\right)\tau^\uparrow_0 - h_x \cos(k/2)\tau^\uparrow_x - h_y \sin(k/2)\tau^\uparrow_y - \left(h_z - \frac{U}{2}\langle\tau^\downarrow_z\rangle_0\right)\tau^\uparrow_z \\ &- \left(\frac{U}{2} - \frac{U}{2}\langle\tau^\uparrow_0\rangle_0\right)\tau^\downarrow_0 - h_x \cos(k/2)\tau^\downarrow_x - h_y \sin(k/2)\tau^\downarrow_y - \left(h_z - \frac{U}{2}\langle\tau^\uparrow_z\rangle_0\right)\tau^\downarrow_z.\end{aligned} \tag{S63}$$

Here, $\tau^{\uparrow/\downarrow}_\alpha$ is a 4×4 matrix defined by $\tau^{\uparrow/\downarrow}_\alpha = (\sigma_0 \pm \sigma_z)/2 \otimes \tau_\alpha$ ($\alpha = 0, x, y, z$), and σ_α and τ_α represent the Pauli matrices (with the identity matrices for $\alpha = 0$) associated with the real spin and sublattice pseudospin, respectively. Equation (S63) indicates that, in the HF approximation, the Hamiltonian reduces to two decoupled Rice–Mele models—one for each spin component—and that only the parameter h_z is renormalized as $h_z \to \tilde{h}^s_z$, where

$$\tilde{h}^\uparrow_z = h_z - \frac{U}{2}\langle\tau^\downarrow_z\rangle_0, \quad \tilde{h}^\downarrow_z = h_z - \frac{U}{2}\langle\tau^\uparrow_z\rangle_0. \tag{S64}$$

Therefore, when the deviation of $\tilde{h}^s_z$ from h_z is small, the second-order response function $\chi^{(2)}(\omega_1, \omega_2)$ remains almost unchanged. In contrast, a sign reversal in $\tilde{h}^s_z$ leads to a global sign flip in $\chi^{(2)}(\omega_1, \omega_2)$ across the entire frequency space.

In Fig. S1, we show the calculated $\tilde{h}^s_z$ in the HF approximation, along with the electron density n_j, as functions of the interaction strength U. In the region $U \le 0.2$ discussed in the main text, the electron density in the HF approximation [Fig. S1(a)] agrees well with the exact iTEBD results. In this regime, the values of $\tilde{h}^s_z$ shown in Fig. S1(b) are only slightly smaller than the $U = 0$ value of $\tilde{h}^s_z = h_z = 0.1$, and they are identical for both spin-up and spin-down components. Specifically, for $U = 0.2$, we find $\tilde{h}^\uparrow_z = \tilde{h}^\downarrow_z = 0.088$ and confirm that $\chi^{(2)}(\omega_1, \omega_2)$ remains qualitatively unchanged from the $U = 0$ case. These results strongly suggest that, even though the ground states for $U \lesssim 0.2$ are well approximated by the HF theory, the observed frequency-dependent sign reversal in $\chi^{(2)}(\omega_1, \omega_2)$ (Fig. 4) is a consequence of many-body effects that are not captured by one-body physics.

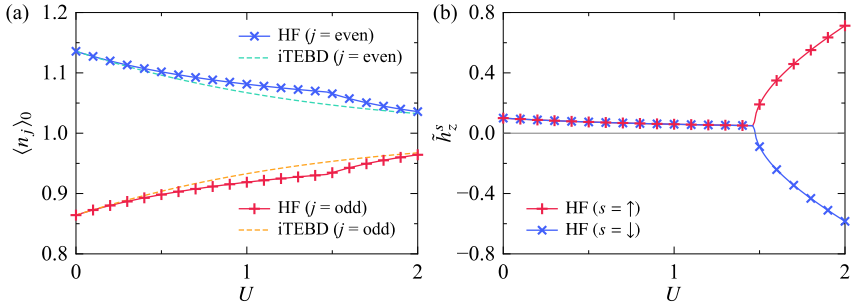


FIG. S1. (a) Electron density $n_j = n_{j\uparrow} + n_{j\downarrow}$ calculated with the HF approximation (solid lines) and the iTEBD with $D = 400$ (dashed lines). (b) Renormalized model parameter $\tilde{h}^s_z$ in the HF approximation. The system undergoes a second-order phase transition from a nonmagnetic ferroelectric phase to an antiferromagnetic ferroelectric phase at $U \approx 1.47$. The parameters are set to $h_y = 0.5$ and $h_z = 0.1$ in (a) and (b).